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Inventor(s)	Suk; Michael et al.

Fluid accelerator

Abstract

A turbine wheel may include an inner hub having a central axis, an outer ring concentric with the inner hub, and an intermediate ring concentrically disposed between the inner hub and the outer ring, and a plurality of turbine blades extending between the intermediate ring and the outer ring. The turbine blades may be oriented to be driven by a fluid stream flowing through the turbine wheel to rotate the turbine wheel in a first direction about the central axis. In addition, the turbine wheel may further include a plurality of compressor blades extending between the inner hub and the intermediate ring. Also, the compressor blades may be oriented to propel a fluid in a downstream direction when the turbine wheel is rotated in the first direction.

Inventors: **Suk; Michael (Milton, PA), Shoffler; David A. (Marion Heights, PA)**

Applicant: **ESS 2 Tech, LLC** (Bethlehem, PA)

Family ID: **92461173**

Assignee: **ESS 2 Tech, LLC (Bethlehem, PA)**

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Primary Examiner: Zamora Alvarez; Eric J

Attorney, Agent or Firm: Plumsea Law Group, LLC

Background/Summary

BACKGROUND OF THE INVENTION

(1) The present invention relates generally to a fluid accelerator and, more particularly, to a fluid accelerator including an airfoil-shaped annular ring, a converging nozzle, and a bi-directional turbine wheel.

(2) Various fluid systems utilize the energy from flowing fluid. For example, wind turbines and ram air turbines convert energy from air streams into electrical and/or hydraulic power. In addition, other fluid systems produce thrust by applying energy to fluid streams to accelerate the fluid stream. For example, jet engines burn fuel to accelerate intake air in order to generate thrust. Both types of systems are limited in their efficiency and effectiveness by the speed of the fluid stream at the inlet side of the fluid system. It would be beneficial to accelerate the fluid stream at the inlet side of either type of system. However, most fluid accelerating devices require power to introduce energy to the fluid stream. For a system that is intended to extract energy from the fluid stream, efficiency is greatly reduced if power is required to accelerate the fluid stream prior to the inlet of the fluid system. Similarly, for systems that already use power to accelerate a fluid stream, adding a separate device that also, itself requires power, is not generally an efficient approach.

(3) There is a need in the art for a system and method that addresses the shortcomings discussed above.

SUMMARY OF THE INVENTION

(4) The present disclosure is directed to a fluid accelerator that is passive. The passive air accelerator greatly increases the efficiency and effectiveness of energy producing systems, such as a ram air turbine, and greatly increases the output and efficiency of energy consumption systems,

such as a jet engine.

(5) In one aspect, the present disclosure is directed to a fluid accelerator. The fluid accelerator may include an outer housing having an inlet end and an outlet end, the outer housing defining a converging nozzle proximate the inlet end. The fluid accelerator may also include an annular ring disposed proximate the inlet end of the outer housing within the converging nozzle, wherein the annular ring has an airfoil cross-sectional shape.

(6) In another aspect, the present disclosure is directed to a fluid accelerator. The fluid accelerator may include an outer housing having an inlet end and an outlet end, the outer housing defining a nozzle proximate the inlet end; and an annular ring disposed proximate the inlet end of the outer housing within the nozzle. The annular ring may have an airfoil cross-sectional shape. In addition, the annular ring may have a leading edge, a trailing edge, a suction side, and a pressure side. The airfoil cross-sectional shape may include a base portion including a first surface associated with the pressure side and a second surface associated with the suction side; an overhang portion that extends over some of the base portion; and an elliptic portion connecting the base portion and the overhang portion adjacent the leading edge. The overhang portion may be curved toward the second surface of the base portion.

(7) In another aspect, the present disclosure is directed to a fluid accelerator. The fluid accelerator may include an outer housing having an inlet end and an outlet end, the outer housing defining a nozzle proximate the inlet end; and an annular ring disposed proximate the inlet end of the outer housing within the nozzle. The annular ring may have an airfoil cross-sectional shape, and may have a suction side of the airfoil cross-sectional shape and a pressure side of the airfoil cross-sectional shape. The annular ring may be angled with the suction side of the airfoil cross-sectional shape being tilted downstream toward the outlet end and the pressure side of the airfoil cross-sectional shape being tilted upstream toward the inlet end.

(8) In another aspect, the present disclosure is directed to a turbine wheel. The turbine wheel may include an inner hub having a central axis, an outer ring concentric with the inner hub, and an intermediate ring concentrically disposed between the inner hub and the outer ring. The turbine wheel may also include a plurality of turbine blades extending between the intermediate ring and the outer ring, the turbine blades being oriented to be driven by a fluid stream flowing through the turbine wheel to rotate the turbine wheel in a first direction about the central axis. In addition, the turbine wheel may include a plurality of compressor blades extending between the inner hub and the intermediate ring, the compressor blades being oriented to propel a fluid in a downstream direction when the turbine wheel is rotated in the first direction.

(9) In another aspect, the present disclosure is directed to a turbine wheel. The turbine wheel may include a plurality of turbine blades oriented to be driven by a fluid stream flowing through the turbine wheel to rotate the turbine wheel in a first direction about the central axis. The turbine wheel may also include a plurality of compressor blades oriented to propel a fluid in a downstream direction when the turbine wheel is rotated in the first direction. The turbine blades and/or the compressor blades may have an airfoil cross-sectional shape. The airfoil cross-sectional shape may include a base portion including a first surface associated with the pressure side and a second surface associated with the suction side; an overhang portion that extends over some of the base portion; and an elliptic portion connecting the base portion and the overhang portion adjacent the leading edge. The overhang portion may be curved toward the second surface of the base portion.

(10) In another aspect, the present disclosure is directed to a fluid accelerator. The fluid accelerator may include an outer housing having an inlet end and an outlet end, the outer housing defining a nozzle proximate the inlet end. In addition, the fluid accelerator may include an annular ring disposed proximate the inlet end of the outer housing within the nozzle, wherein the annular ring has an airfoil cross-sectional shape. Also, the fluid accelerator may include a turbine wheel disposed within the outer housing downstream from the annular ring. The turbine wheel may include an inner hub having a central axis, an outer ring concentric with the inner hub, and an

intermediate ring concentrically disposed between the inner hub and the outer ring. The turbine wheel may also include a plurality of turbine blades extending between the intermediate ring and the outer ring, the turbine blades being oriented to be driven by a fluid stream through the turbine wheel to rotate the turbine wheel in a first direction about the central axis. In addition, the turbine wheel may include a plurality of compressor blades extending between the inner hub and the intermediate ring, the compressor blades being oriented to drive a fluid downstream when the turbine wheel is rotated in the first direction.
(11) Other systems, methods, features, and advantages of the invention will be, or will become, apparent to one of ordinary skill in the art upon examination of the following figures and detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description and this summary, be within the scope of the invention, and be protected by the following claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The invention can be better understood with reference to the following drawings and description. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating the principles of the invention. Moreover, in the figures, like reference numerals designate corresponding parts throughout the different views.
(2) FIG. **1** is a schematic isometric view of an embodiment of an airfoil along the pressure side;
(3) FIG. **2** is a schematic isometric view of a suction side of the airfoil of FIG. **1**;
(4) FIG. **3** is a schematic side view of an embodiment of an airfoil;
(5) FIG. **4** is a schematic side view of the airfoil of FIG. **3**, in which the curvature of various portions of the airfoil are indicated;
(6) FIG. **5** is a schematic view of an embodiment of an airfoil indicating pathlines of airflow elements;
(7) FIG. **6** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system;
(8) FIG. **7** is a schematic illustration of the inside of an aircraft body shell showing components of a ram air turbine system;
(9) FIG. **8** is a schematic rear view of a ram air turbine according to an exemplary embodiment;
(10) FIG. **9** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system according to an alternative embodiment;
(11) FIG. **10** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system according to another alternative embodiment;
(12) FIG. **11** is a schematic illustration of a front perspective view of a fluid accelerator according to an exemplary embodiment;
(13) FIG. **12** is a schematic illustration of a rear perspective view of the fluid accelerator shown in FIG. **11**;
(14) FIG. **13** is a schematic illustration of a rear perspective exploded view of the fluid accelerator shown in FIG. **11**;
(15) FIG. **14** is a schematic illustration of a perspective cutaway cross-sectional view of a nozzle of the fluid accelerator shown in FIG. **11**;
(16) FIG. **15** is a schematic illustration of a perspective cutaway cross-sectional view of a nozzle and turbine wheel of the fluid accelerator shown in FIG. **11**;
(17) FIG. **16** is a schematic illustration of a perspective cutaway cross-sectional view of the fluid accelerator shown in FIG. **11**;
(18) FIG. **17** is a schematic illustration of a perspective cutaway cross-sectional view of the outer

housing and annular ring of the fluid accelerator shown in FIG. **11** shown with fluid flow patterns effectuated by the outer housing and annular ring;

(19) FIG. **18** is a schematic illustration of a cutaway cross-sectional view of the turbine wheel of the fluid accelerator shown in FIG. **11**;

(20) FIG. **19** is a schematic illustration of an airplane having a fluid accelerator mounted at the inlet of its turbine engines.

DETAILED DESCRIPTION

(21) The present disclosure is directed to systems that implement airfoils having hooked cross-sectional shapes. The airfoils of the disclosed embodiments may implement one or more airfoil shapes described in Suk et al., U.S. Pat. No. 10,766,544, issued on Sep. 8, 2020, and entitled “Airfoils and Machines Incorporating Airfoils,” the entire disclosure of which is incorporated herein by reference.

(22) In addition, the present disclosure is directed to systems that are usable with an internally mounted ram air turbine system. Accordingly, the disclosed systems may be used with a ram air turbine system having one or more features described in Suk et al., U.S. patent application Ser. No. 18/173,748, filed on Feb. 23, 2023, and entitled “Ram Air Turbine,” the entire disclosure of which is incorporated herein by reference.

(23) As used herein, the term “airfoil” (or “aerofoil”) is any structure with curved surfaces that produces an aerodynamic force when moved through a fluid. As used herein, the term “fluid” may refer to any Newtonian Fluid. In other embodiments, airfoils could be used with Non-Newtonian Fluids.

(24) An airfoil may include an upper or suction surface against which fluid flows at a relatively high velocity with low static pressure. An airfoil may also include a lower or pressure surface that has a high static pressure relative to the suction surface. Alternatively, the suction and pressure surfaces could be referred to as suction and pressure sides. The airfoil also includes a leading edge defined as the point at the front of the airfoil with maximum curvature. The airfoil also includes a trailing edge defined as the point at the rear of the airfoil with minimum curvature. In addition, a chord line of the airfoil refers to a straight line between the leading and trailing edges. Also, a mean camber line is the locus of points midway between the upper and lower surfaces and may or may not correspond with the chord line depending on the shape of the airfoil.

(25) As used herein, an airfoil has a chord length defined as the length of the airfoil's chord line. In addition, the airfoil has a thickness defined as the distance between the upper and lower surfaces along a line perpendicular to the mean camber line. The width of an airfoil is taken in a direction perpendicular to both the chord line and the thickness.

(26) Throughout the specification and claims the term “radius of curvature” is used. The radius of curvature is the reciprocal of the curvature at a particular location on a curve or two-dimensional surface. For a curve, the radius of curvature equals the radius of the circular arc that best approximates the curve at that point. In particular, it should be noted that the larger the radius of curvature of a curve, the smaller the curvature (and vice versa).

(27) FIGS. **1-2** illustrate schematic isometric views of an airfoil (or blade) **100**, while FIG. **3** illustrates a schematic side view of airfoil **100**. Referring to FIGS. **1-3**, airfoil **100** is comprised of a pressure side **102** (shown in FIG. **1**) and an opposing suction side **104** (shown in FIG. **2**). Each of these sides includes a surface (e.g., a pressure surface or a suction surface) in contact with air during operation. Additionally, airfoil **100** includes a leading edge **110** and a trailing edge **112**. Moreover, leading edge **110** and trailing edge **112** are connected by a chord line **114** (see FIG. **3**).

(28) Referring to FIG. **3**, in some embodiments, airfoil **100** may be comprised of a single body **101** of material. In some embodiments, airfoil **100** may be formed of a single unitary piece of material. Starting from trailing edge **112**, body **101** includes a base portion **180** that is seen to curve gradually through the length of airfoil **100**. Base portion **180** includes first a surface **187** on pressure side **102** of airfoil **100** and an opposing second surface **188** on opposing suction side **104**.

(29) At the end of base portion **180**, body **101** bends to create a folded or hook-like section adjacent leading edge **110**. That is, adjacent leading edge **110**, body **101** is comprised of an elliptic portion **142** as well as an overhang portion **182** that hangs or extends over some of base portion **180**. Elliptic portion **142** connects base portion **180** and overhang portion **182** and also includes leading edge **110**.

(30) In some embodiments, overhang portion **182** may be spaced apart or separated from base portion **180**. In the embodiment of FIG. **3**, overhang portion **182** is separated from base portion **180** by gap **195**. In different embodiments, the size of gap **195** may vary. In some cases, gap **195** may be greater than or equal to a thickness of overhang portion **182**. In some cases, gap **195** may be at least three times as large as a thickness of overhang portion **182**. The size of gap **195**, and thus, the spacing of overhang portion **182** from base portion **180**, determines the airflow behavior across airfoil **100**. Accordingly, as discussed in further detail below, gap **195** may be sized accordingly to provide the desirable airflow behavior.

(31) The fold in body **101** adjacent leading edge **110** may be seen to divide airfoil **100** into two portions having distinctive geometries: a leading airfoil portion **120** and a trailing airfoil portion **122**. As shown in FIG. **3**, leading airfoil portion **120** is formed by the elliptic portion **142**, the overhang portion **182**, and the leading segment of base portion **180**. Trailing airfoil portion **122** is formed by the trailing segment of base portion **180**.

(32) In different embodiments, the length of leading airfoil portion relative to the overall length of the airfoil (that is, the percent of the total airfoil length that the overhang portion extends over) can vary. In some cases, the leading airfoil portion has a relative length of 25 to 50 percent of the total airfoil length. In one embodiment, the leading airfoil portion has a length of at least 25 percent of the total airfoil length. In yet another embodiment, the leading airfoil portion has a length of at least one third of the total airfoil length. In some cases, the leading airfoil portion may made sufficiently long enough (at least 25 percent or so of the total airfoil length) so that the first arc portion can be gradually curved down towards the second arc portion, thereby helping to keep the boundary layer attached to the airfoil before the dramatic step down in thickness adjacent the second arc portion.

(33) As seen in FIG. **3**, body **101** has a relatively constant local thickness **103** throughout airfoil **100**. However, the folded shape of body **101** that forms overhang portion **182** provides a greater overall thickness through leading airfoil portion **120** than in trailing airfoil portion **122**. Here, the overall thickness is measured between opposing suction side **104** and pressure side **102** and is distinct from the local body thickness. Specifically, leading airfoil portion **120** has variable thickness **130** with a maximum value adjacent leading edge **110** and a minimum value at a location furthest from leading edge **110**. In contrast, trailing airfoil portion **122** has an approximately constant thickness. In some embodiments, the thickness of trailing airfoil portion **122** is approximately equal to local thickness **103** of body **101**. In other embodiments, trailing airfoil portion **122** could also have a variable thickness.

(34) An airfoil may include provisions for keeping airflow “stuck” on the suction surface so that the air can be redirected through a large angle (e.g., from a near horizontal direction for incoming air to a near vertical direction for outgoing air). In some embodiments, an airfoil can include a leading airfoil portion that includes one or more arcs for controlling the flow of air along a suction surface.

(35) In some embodiments, overhang portion **182** may be further comprised of a first arc portion **152** and a second arc portion **154**. First arc portion **152** may extend from elliptic portion **142**, while second arc portion **154** may be disposed at an open or free end of overhang portion **182**. In some embodiments, the curvature (along opposing suction side **104**) of overhang portion **182** may vary from first arc portion **152** to second arc portion **154**. In some cases, first arc portion **152** may be configured to curve down in the direction of base portion **180**. Moreover, second arc portion **154** may be configured with steeper curvature that is also directed downwardly toward base portion **180**.

(36) In the following description the radius of curvature of various surfaces is defined relative to

the length of a unit radius, denoted as “UN.” In different embodiments, the particular value of the length of the unit radius could vary. For example, the unit radius could have a length of 100 mm (i.e., 1 UN=100 mm), 6 inches (i.e., 1 UN=6 inches), or any other value. It may be understood that the ratio of two radii of curvature is independent of the particular value of the unit radius. Thus, if a first surface has a radius of curvature of 1 UN and a second surface has a radius of curvature of 0.5 UN, the ratio is equal to 1 divided by 0.5, or 2, and is a dimensionless quantity that is independent of the particular length of the unit radius in a given embodiment.

(37) FIG. **4** is a schematic side view of airfoil **100**. Referring to FIG. **4**, the different portions or segments of airfoil **100** can have different degrees of curvature. In some embodiments, trailing airfoil portion **122** has trailing arc portion **160** immediately adjacent trailing edge **112**. Trailing arc portion **160** has a radius of curvature **200**. In some cases, radius of curvature **200** could have a value of approximately 0.1000 UN. In some embodiments, main segment **162** of trailing airfoil portion **122** has a radius of curvature **202** along pressure side **102** and a radius of curvature **204** along opposing suction side **104**. In some cases, radius of curvature **202** has a value of approximately 1.1250 UN. In some cases, radius of curvature **204** has a value of approximately 1.1750 UN. In some embodiments, elliptic portion **142** has a radius of curvature **206** on outward facing side **170** and a radius of curvature **208** on inward facing side **172**. In some cases, radius of curvature **206** has a value of approximately 0.1500 UN. In some cases, radius of curvature **208** has a value of approximately 0.1000 UN.

(38) In some embodiments, first arc portion **152** has a radius of curvature **210** along opposing suction side **104** and a radius of curvature **212** along inward facing surface **190**. In some cases, radius of curvature **210** has a value of approximately 0.7500 UN. In some cases, radius of curvature **212** has a value of approximately 0.7000 UN. In addition, second arc portion **154** has a radius of curvature **214**. In some cases, radius of curvature **214** has a value of approximately 0.1000 UN.

(39) In some embodiments, the curvature of each segment of airfoil **100** may be selected to help keep the boundary layer of flowing air attached to opposing suction side **104**, even as airfoil **100** curves from leading edge **110** to trailing edge **112**.

(40) FIG. **5** is a schematic view of airfoil **100** in operation as air passes across it. Referring to FIG. **5**, incoming air flows in a first direction **300** and encounters leading edge **110** first. Air moving across opposing suction side **104** will first pass across first arc portion **152**, which curves to second arc portion **154**. Air then gets directed into trailing airfoil portion **122**. As the air flows along trailing airfoil portion **122**, it is directed to trailing arc portion **160** and turns as it leaves trailing edge **112**.

(41) The geometry of leading airfoil portion **120** creates step-down region **310** resulting in an abrupt change in thickness between leading airfoil portion **120** and trailing airfoil portion **122**. This sudden change in thickness (and geometry) creates vortex **320** (and/or turbulent eddies) at step-down region **310**. As air flows over opposing suction side **104**, vortex **320** “pulls” the air toward the airfoil and thereby reattaches the boundary layer of the flow as it moves from one section of the airfoil to the next, keeping the air “stuck” on opposing suction side **104**.

(42) The embodiments utilize specifically curved arc portions adjacent step-down region **310** to help actively control the turbulent eddies or vortices that develop at step-down region **310**. Specifically, first arc portion **152** and second arc portion **154** combine to actively redirect the fluid flow with use of the Coandă effect toward reattachment to the airfoil surface. The Coandă effect refers to the tendency of a jet of fluid emerging from an orifice to follow an adjacent flat or curved surface and to entrain fluid from the surroundings so that a region of lower pressure develops. Vortex **320** (and/or turbulent eddies) at step-down region **310** creates a pressure difference between second arc portion **154** and trailing airfoil portion **122**. The active fluid flowing across opposing suction side **104** creates an air curtain **322** (via the Coandă effect) that helps hold vortex **320** in place and keeps it attached to opposing suction side **104**. Air curtain **322** thus provides a stabilizing force to keep vortex **320** in place, which further serves to prevent the boundary layer from

delaminating from airfoil **100**.
(43) This arrangement provides an airfoil that keeps the airflow stuck to opposing suction side **104** enough to turn the airflow direction by close to 90 degrees. That is, air initially flowing in first direction **300** as it encounters leading edge **110** leaves trailing edge **112** traveling in a second direction **302**. In some cases, second direction **302** may be approximately 90 degrees From first direction **300**. Thus, the airfoil will be forced in a direction that is substantially opposite of first direction **302**. When the airfoil is included in a rotational turbine blade set, this redirection of airflow, resulting in opposing forces that drive the airfoil in a direction that is substantially 90 degrees From the direction of air flow into which the airfoil is placed, may maximize the amount of rotational energy to which the airflow is converted. This may maximize the amount of electrical power that may be generated by such a turbine blade set. In other embodiments, depending on the shape and local curvature of various segments of airfoil **100**, the direction of incoming air could be changed by any amount between approximately 10 and 90 degrees.
(44) In different embodiments, the disclosed airfoils could be manufactured from various materials. Exemplary materials include, but are not limited to, materials known for use in manufacturing turbine blades (e.g., U-500, Rene 77, Rene N5, Rene N6, PWA1484, CMSX-4, CMSX-10, Inconel, GTD-111, EPM-102, Nominic 80a, Niminic 90, Nimonic 105, Nimonic 105 and Nimonic 263). Other materials include ceramic matrix composites. Other materials for airfoils can include, but are not limited to, aluminum, composite materials, steel, titanium as well as other materials.
(45) Airfoils can be manufactured using any known methods. In some embodiments, an airfoil can be formed using an extrusion process.
(46) The dimensions of an airfoil can vary according to its intended application. The chord line length, width, and thickness can all be varied in different ratios while maintaining the general profile shape of the airfoil.
(47) A turbine blade set that implements blades having the airfoil shape described above may be utilized in a ram air turbine. Such a ram air turbine may be configured for use in an aircraft. In some embodiments, the ram air turbine may be fixedly housed inside the body shell of the aircraft. Conduits may direct air from the air stream around the aircraft into the body shell to the internal ram air turbine and back out of the body shell.
(48) FIG. **6** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system. As shown in FIG. **6**, an aircraft **600** may include a ram air turbine system **602**. Aircraft **600** may include a body shell **605**. System **602** may be located at any suitable location of aircraft **600**. For example, in some embodiments, system **602** may be located in the fuselage. In other embodiments, system **602** may be located on a wing. Further, in some embodiments, system **602** may be located in a flap track fairing, as shown in FIG. **6**. Air flow adjacent the flap track fairing is typically smooth, and thus, desirable for placement of a ram air turbine system because it produces consistent power, which is favorable for producing stable amounts of hydraulic pressure. It will be understood, however, that other locations may be suitable for the disclosed ram air turbine system.
(49) As shown in FIG. **6**, system **602** may include an air inlet **610**, into which air from the air stream outside of body shell **605** may flow, as indicated by an arrow **625**. Air entering air inlet **610** may be directed to a ram air turbine located inside body shell **605** (see FIG. **7**). Air inlet **610** may have any suitable shape to enable smooth flow of air to be delivered to the ram air turbine. Bevels, curves, and other shapes may be used for the surfaces in and around air inlet **610** to affect aerodynamics of the air flow delivered to the ram air turbine. The size of air inlet **610** may be suited to provide air flow to the ram air turbine with the volume, speed, and consistency desired for operation of the ram air turbine.
(50) In some embodiments, additional airflow may be directed into air inlet **610** by an air scoop **620**. In some embodiments air scoop **620** may be deployable. Accordingly, in an undeployed position (not shown), air scoop **620** may sit flush, or substantially flush, with the outer surface of

body shell **605**. In a deployed position, as shown in FIG. **6**, air scoop **620** may collect and guide or direct air into air inlet **610**. Air scoop **620** may include suitable features to provide air flow at a speed and consistency desired for operation of the ram air turbine. Air scoop **620** may be formed of any suitable material, such as carbon fiber, aluminum, titanium, steel, stainless steel, or any other material with the strength and rigidity to be deployed into the airstream around the aircraft on which it is implemented.

(51) The deployment mechanism of air scoop **620** is not shown. It will be understood that any suitable deployment mechanism may be used for the deployment of air scoop **620**, and that skilled artisans will appreciate mechanisms that are suitable for deploying an air scoop into an air stream at a given aircraft speed. Because of the simplicity of air scoop **620** compared to any ram air turbine, the deployment mechanism for air scoop **620** may have a reduced size, weight, and/or complexity compared to deployment mechanisms for ram air turbines.

(52) In addition, as also shown in FIG. **6**, body shell **605** may include an air outlet **615**. Air exiting the ram air turbine within body shell **605** may be directed out of air outlet **615**. Accordingly, to facilitate the air flow through the ram air turbine, air inlet **610** and air outlet **615** may be arranged spaced from one another in the direction of aircraft flight. While the two ports need not necessarily be located directly in line with one another, the air flow through the ram air turbine is facilitated if air inlet **610** is positioned further forward toward the front of aircraft **600** and air outlet **615** is positioned further aft toward the rear of aircraft **600**.

(53) Although not shown in FIG. **6**, in some embodiments, air outlet **615** may include a closure plate or other structure to close it off when the ram air turbine is not in use. In some cases, operation of the closure plate may be mechanically linked to the deployment of air scoop **620**. Other mechanisms to effectuate the opening of the closure plate when needed may also be utilized, such as spring-bias, air pressure, etc.

(54) FIG. **7** is a schematic illustration of the inside of aircraft body shell **605** showing components of ram air turbine system **602**. As shown in FIG. **7**, system **602** may include an intake conduit **611** configured to guide air from air inlet **610** in body shell **605** to a ram air turbine, which may be housed in a ram air turbine housing **700**. In addition, system **602** may include an outlet conduit **616** configured to guide air exiting the ram air turbine to air outlet **615** in body shell **605**. The shape, size, and material with which air intake conduit **611** and outlet conduit **616** are formed may be selected accordingly to deliver air flow to the ram air turbine in the amount, speed, and smoothness desired.

(55) Thus, in some embodiments, the ram air turbine may be fixedly housed inside body shell **605** of the aircraft. In addition, an electric generator may be associated with, and configured to be driven by, the ram air turbine. Accordingly, a generator housing **705** may be configured to fixedly house the generator inside body shell **605** of the aircraft.

(56) By keeping the ram air turbine inside the aircraft, and only deploying an air scoop, not only may weight be saved because the deployment mechanism for the air scoop is lighter than that of a ram air turbine, but also the deployed air scoop may produce less drag than a ram air turbine that is, itself, deployed into the air stream.

(57) FIG. **8** is a schematic rear view of a ram air turbine **800** according to an exemplary embodiment. As shown in FIG. **8**, ram air turbine **800** may have an upstream end **805** and a downstream end **810**. Thus, ram air turbine **800** may be configured to receive air flowing through ram air turbine **800** from upstream end **805** to downstream end **810**.

(58) In addition, ram air turbine **800** may include an outer housing **820**. In some embodiments, outer housing may be tapered. For example, as shown in FIG. **8**, outer housing **820** may have a larger diameter at downstream end **810** than at upstream end **805**.

(59) FIG. **8** also shows an output shaft **815** configured to be driven by turbine blade sets **825** within outer housing **820**. Output shaft **815** may ultimately drive an electric power generator, a hydraulic pump, or both.

(60) FIG. **9** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system according to an alternative embodiment. As shown in FIG. **9**, an aircraft may include a ram air turbine system **902**. The aircraft may include a body shell **905**. Like system **602**, system **902** may be located at any suitable location of the aircraft.
(61) As shown in FIG. **9**, system **902** may include an air inlet **910**, into which air from the air stream outside of body shell **905** may flow, as indicated by an arrow **925**. Air entering air inlet **910** may be directed to a ram air turbine located inside body shell **905** (see FIG. **7**). Additional airflow may be directed into air inlet **910** by an air scoop **920**. In some embodiments air scoop **920** may be deployable. Accordingly, in an undeployed position (not shown), air scoop **920** may sit flush, or substantially flush, with the outer surface of body shell **905**. In a deployed position, as shown in FIG. **9**, air scoop **920** may collect and guide or direct air into air inlet **910**. Air scoop **920** may include suitable features to provide air flow at a speed and consistency desired for operation of the ram air turbine. Air scoop **920** may be formed of any suitable material, such as carbon fiber, aluminum, titanium, steel, stainless steel, or any other material with the strength and rigidity to be deployed into the airstream around the aircraft on which it is implemented.
(62) The deployment mechanism of air scoop **920** is not shown. It will be understood that any suitable deployment mechanism may be used for the deployment of air scoop **920**, and that skilled artisans will appreciate mechanisms that are suitable for deploying an air scoop into an air stream at a given aircraft speed.
(63) In addition, as also shown in FIG. **9**, body shell **905** may include an air outlet **915**. Air exiting the ram air turbine within body shell **905** may be directed out of air outlet **915**. Accordingly, to facilitate the air flow through the ram air turbine, air inlet **910** and air outlet **915** may be arranged spaced from one another in the direction of aircraft flight. While the two ports need not necessarily be located directly in line with one another, the air flow through the ram air turbine is facilitated if air inlet **910** is positioned further forward toward the front of aircraft **900** and air outlet **915** is positioned further aft toward the rear of aircraft **900**.
(64) In addition to the air stream flowing into inlet **910** and through the RAT inside the aircraft body shell **905**, airflow through the system is also driven by the venturi effect at outlet **915**. While the venturi effect does drive the air flow through system **602**, this effect may be tempered by the presence of air scoop **620** being aligned with outlet **615** in the direction of the air stream. In order to maximize the venturi effect, in some embodiments, the outlet may be located not only spaced in the direction of air flow, but also laterally. For example, as shown in FIG. **9**, outlet **915** may be spaced laterally from inlet **910**, as indicated by an arrow **935**. Accordingly, a clean airstream flows over outlet **915**, undisrupted by air scoop **920**, as indicated by an arrow **940**.
(65) In some embodiments, the internal RAT system can also include a fluid accelerator at the inlet, to increase the speed of air delivered to the RAT. Since the accelerator discussed herein works in liquids as well as gases, it is referred to as a “fluid” accelerator. In air applications, such as an RAT, it may also be referred to as an “air” accelerator. FIG. **10** is a schematic illustration of an aircraft with an enlarged view of externally exposed components of a ram air turbine system according to another alternative embodiment. As shown in FIG. **10**, an aircraft **1000** may include an internal ram air turbine system **1002**. System **1002** may include an air inlet **1010** and an air outlet **1015**. System **1002** may also include an air scoop **1020** configured to guide air into air inlet **1010**, as indicated by arrow **1025**, with air exiting air outlet **1015** being indicated by arrow **1030**. As shown in FIG. **10**, air outlet **1015** may be laterally spaced from air inlet **1010**, as indicated by arrow **1035**, so that an undisturbed airstream, indicated by arrow **1040**, flows over air outlet **1015** to produce a venturi effect.
(66) Other features of system **1002** may be the same or similar to system **602** and **902** discussed above. In addition, system **1002** may include a fluid accelerator **1050** disposed at inlet **1010**. Any suitable fluid accelerator may be utilized with system **1002**. Exemplary features of such a fluid accelerator are discussed below in greater detail.

(67) FIG. **11** is a schematic illustration of a front perspective view of a fluid accelerator according to an exemplary embodiment. As shown in FIG. **11**, a fluid accelerator **1100** may include an outer housing **1105** having an inlet end **1110** and an outlet end **1115**. When fluid flows from inlet end **1110** to outlet end **1115**, the fluid is accelerated.
(68) Outer housing **1105** may be formed of any suitable material, such as metals, plastics, composites, and or other construction materials. In addition, although outer housing **1105** is shown as having a circular cross-section, certain portions of outer housing **1105** may have a different cross-sectional shape.
(69) Multiple fluid accelerating components may be disposed within outer housing **1105**. For example, as shown in FIG. **11**, fluid accelerator **1100** may include an annular ring **1130**, which may have an airfoil cross-sectional shape. In addition, fluid accelerator **1100** may also include a turbine wheel **1150**, including both turbine blades and compressor blades.
(70) As shown in FIG. **11**, outer housing **1105** may include an inlet portion **1120** and an outlet portion **1125**. In some embodiments, outer housing **1105** may define a converging nozzle proximate inlet end **1110**. For example, as shown in FIG. **11**, inlet portion **1120** of outer housing **1105** may be a converging nozzle. That is, as shown in FIG. **11**, inlet portion **1120** may be substantially conical, with a diameter that reduces along the direction of fluid flow. Within the converging nozzle of inlet portion **1120** of outer housing **1105**, annular ring **1130** may be disposed proximate inlet end **1110**.
(71) As shown in FIG. **11**, annular ring **1130** may be disposed proximate inlet end **1110** of outer housing **1105** within the converging nozzle of outer housing **1105**. As also shown in FIG. **11**, annular ring **1130** may be spaced from outer housing **1105** by a plurality of spacer mounts **1135**, as indicated by an arrow **1140**. Any suitable number of spacer mounts **1135** may be used to support annular ring **1130**. For example, more flexible and/or delicate annular rings may require more spacer mounts, whereas more robust annular rings may be provided with fewer spacer mounts. The spacing between annular ring **1130** and outer housing **1105** indicated by arrow **1140** may be determined based on the use of fluid accelerator **1100**. This spacing, along with other various parameters such as the shape of the airfoil, size of the airfoil, angle of the converging nozzle, etc., may be selected according to the intended use of the device. Factors for determining these parameters may include the type of fluid (e.g., air, gas, liquid, etc.), speed of fluid flow entering the fluid accelerator, desired speed of fluid flow exiting the fluid accelerator, size and fitment constraints, etc.
(72) FIG. **12** is a schematic illustration of a rear perspective view of the downstream end of the fluid accelerator shown in FIG. **11**. As shown in FIG. **12**, fluid accelerator **1100** may also include a mounting cross-member **1155** for turbine wheel **1150**. As shown in FIG. **12**, mounting cross-member **1155** may include a peripheral ring **1160**, which may be mounted to outer housing **1105**. In addition, mounting cross-member **1155** may have a central hub **1170**, which is connected to peripheral ring **1160** by a plurality of supports **1165**. Central hub **1170** may have a center hole **1175** configured to receive a connecting shaft **1300** (See FIG. **13**.). Connecting shaft **1300** may be received within center hole **1175** and a center opening **1180** of turbine wheel **1150** in order to mount turbine wheel **1150** within outer housing **1105**. As mounted, turbine wheel **1150** is free to rotate within outer housing **1105**, while mounting cross-member **1155** is fixed with respect to outer housing **1105**.
(73) FIG. **13** is a schematic illustration of a rear perspective exploded view of the fluid accelerator shown in FIG. **11**. As shown in FIG. **13**, a connecting shaft **1300** may be used to mount turbine wheel **1150** to mounting cross-brace **1155**. Connecting shaft **1300** is illustrated generically in FIG. **13**. It will be understood that connecting shaft **1300** may have any configuration suitable to rotatably connect turbine wheel **1150** with mounting cross-brace **1155**. In order to permit turbine wheel **1150** to rotate with respect to mounting cross-brace **1155**, one or more bushings, bearings, or other rotatable connecting devices may be provided in center opening **1180** of turbine wheel **1150**, in center hole **1175** of mounting cross-brace **1155**, at the upstream end of connecting shaft **1300**,

and/or at the downstream end of connecting shaft **1300**.

(74) Annular ring **1420** may have any suitable airfoil shape wherein the suction side faces generally radially outward and is tilted downstream, and the pressure side is facing radially inward and is tiled upstream. In some embodiments, annular ring **1420** may have an airfoil shape configured as described above with respect to FIGS. **1-5**. That is, annular ring **1420** may have a hooked cross-sectional shape.

(75) FIG. **14** is a schematic illustration of a perspective cutaway cross-sectional view of a nozzle of the fluid accelerator shown in FIG. **11**. As shown in FIG. **14**, annular ring **1130** has a leading edge **1400** and a trailing edge **1405**. In addition, annular ring **1130** has a suction side **1420** of the airfoil cross-sectional shape and a pressure side **1425** of the airfoil cross-sectional shape. As further shown in FIG. **14**, annular ring **1130** may be angled with suction side **1420** of the airfoil cross-sectional shape being tilted downstream toward outlet end **1115** of outer housing **1105** and with pressure side **1125** of the airfoil cross-sectional shape being tilted upstream toward inlet end **1110** of outer housing **1105**.

(76) In some embodiments, a line extending through the leading edge and the trailing edge of the annular ring may be substantially parallel to the inner surface of the converging nozzle proximate the inlet end of the outer housing. For example, the tilting of annular ring **1130** is illustrated in FIG. **14** by airfoil axis **1410** extending through leading edge **1400** and trailing edge **1405** of annular ring **1130**. Notably, airfoil axis **1410** is angled with respect to central axis **1430** extending through outer housing **1105**. As shown in FIG. **14**, in some embodiments, airfoil axis **1410** may be substantially parallel to an inner surface **1415** of the converging nozzle formed by inlet portion **1120** of outer housing **1105**.

(77) Also, for reference, FIG. **14** illustrates that, at inlet end **1110**, outer housing **1105** has a first radius **1435**, and in a neck region **1440** between inlet portion **1120** and outlet portion **1125**, outer housing **1105** may have a second radius **1445** that is less than first radius **1435**. Thus, inlet portion **1120** forms a converging nozzle.

(78) In addition to an annular ring having an airfoil cross-sectional shape, the fluid accelerator may include a turbine wheel disposed within the outer housing downstream from the annular ring. The turbine wheel may include a plurality of turbine blades oriented to be driven by a fluid stream flowing through the turbine wheel to rotate the turbine wheel in a first direction about the central axis. In addition, the turbine wheel may include a plurality of compressor blades oriented to propel a fluid in a downstream direction when the turbine wheel is rotated in the first direction. In order to produce this effect, the turbine blades are positioned radially outward of the compressor blades. This way, the torque produced by the turbine blades is higher because the turbine blades are disposed further away from the central axis of the turbine wheel. The larger moment arm of the longer radius produces higher torque.

(79) FIG. **15** is a schematic illustration of a perspective cutaway cross-sectional view of a nozzle and turbine wheel of the fluid accelerator shown in FIG. **11**. As shown in FIG. **15**, turbine wheel **1150** may be disposed within outer housing **1105** downstream from annular ring **1130**. Turbine wheel **1150** may include an inner hub **1500** having a central axis **1430**, an outer ring **1505** concentric with inner hub **1500**, and an intermediate ring **1510** concentrically disposed between inner hub **1500** and outer ring **1505**. In addition, turbine wheel **1150** may include a plurality of turbine blades **1515** extending between intermediate ring **1510** and outer ring **1505**. Turbine blades **1515** may be oriented to be driven by a fluid stream through turbine wheel **1150** to rotate turbine wheel **1150** in a first direction about central axis **1430**, as indicated by an arrow **1525**. In the embodiments illustrated herein, turbine wheel **1150** is configured to be driven in a counter-clockwise direction (when viewed from the upstream side of the turbine wheel) by turbine blades **1515**, as indicated by arrow **1525**. However, it will be understood that, in other embodiments, the turbine wheel may be configured to be driven in a clockwise direction. The orientation of the turbine blades may simply be reversed to drive the turbine wheel in the clockwise direction.

(80) As shown in FIG. **15**, turbine wheel **1150** may also include a plurality of compressor blades **1520** extending between inner hub **1500** and intermediate ring **1510**. Compressor blades **1520** may be oriented to drive a fluid downstream when turbine wheel **1150** is rotated in the first direction (arrow **1525**) by turbine blades **1515**.
(81) FIG. **16** is a schematic illustration of a perspective cutaway cross-sectional view of fluid accelerator **1100**. FIG. **16** illustrates fluid accelerator **1100** with mounting cross-brace **1155** disposed within outer housing **1105**. FIG. **16** does not illustrate fasteners that would be utilized to attach mounting cross-brace **1155** to outer housing **1105**. It will be understood that any suitable type of fasteners, such as bolts, rivets, etc., may be used to fixedly attach or removably attach mounting cross-brace **1155** to outer housing **1105**. In addition, the connecting shaft between turbine wheel **1150** and mounting cross-brace **1155** is also omitted from FIG. **16**. It will be understood that, as discussed above, a connecting shaft will be used to mount turbine wheel **1150** to mounting cross-brace **1155** within outer housing **1105**. (See also FIG. **13**.)
(82) FIG. **17** is a schematic illustration of a perspective cutaway cross-sectional view of outer housing **1105** and annular ring **1130** of fluid accelerator **1100** shown with fluid flow patterns effectuated by outer housing **1105** and annular ring **1130**. There are several ways in which outer housing **1105** and annular ring **1130** accelerate fluid flow through the fluid accelerator.
(83) The airfoil shape of annular ring **1130** accelerates the air flowing over suction side **1420**. This results in a reduction in pressure proximate to suction side **1420** of annular ring **1130**, which produces several effects. In particular, this reduction in pressure on suction side **1420** of annular ring **1130** draws fluid into this area of reduced pressure from several places.
(84) First, this reduced pressure draws fluid from beyond the radius of inlet end **1110** of outer housing **1105**. This is illustrated by a first flow line **1701**. As shown in FIG. **17**, first flow line begins at a larger radius than inlet end **1110**. Absent the reduced pressure on suction side **1420** of annular ring **1130**, the fluid at the beginning of first flow line **1701** would simply bypass around the outer edge of inlet end **1110**. However, this reduced pressure draws this fluid from this circumferential area outside the radius of inlet end **1110**. Thus, a greater amount of fluid is essentially funneled through inlet end **1110** than would be if annular ring **1130** were not disposed within outer housing **1105**. By forcing more fluid through a given size opening, the fluid is accelerated. This is illustrated by the increased length of the dashes of first fluid flow line **1701** as it enters outer housing **1105**.
(85) Second, this reduced pressure on suction side **1420** of annular ring **1130** is located in the area between annular ring **1130** and inlet end **1110** of outer housing **1105**. Accordingly, as illustrated in FIG. **17** by a second fluid flow line **1702**, fluid flowing directly into the space **1140** between annular ring **1130** and inlet end **1110** of outer housing **1105** is accelerated. The increased speed of the fluid is illustrated by the increased length of the dashes of second fluid flow line **1702** as it passes over suction side **1420** of annular ring **1130**.
(86) In addition, because of the tilting of annular ring **1130**, pressure side **1425** of annular ring **1130** forms a converging nozzle. Accordingly, fluid flowing proximate pressure surface **1425** of annular ring **1130** is accelerated. This is illustrated by a third fluid flow line **1703**. As shown in FIG. **17**, the dashes of third fluid flow line **1703** get longer as the fluid passes over pressure side **1425** and converges. The effects of this converging nozzle formed by the tilted pressure side **1425** of annular ring **1130** diminish the further the fluid flow is spaced radially inward from annular ring **1130**. For example, a fourth fluid flow line **1704** illustrates an increase in speed, but the length of the dashes do not increase to the same extent as those of third fluid flow line **1703**. Fifth fluid flow line **1705**, shows less acceleration than fourth fluid flow line **1704**. A sixth fluid flow line **1706** and a seventh fluid flow line **1707** show progressively less acceleration. In addition, an eighth fluid flow line **1708**, which is proximate central axis **1430** of outer housing **1105**, shows the least acceleration.
(87) In addition to the fluid acceleration provided by suction side **1420** of annular ring **1130** and by the converging nozzle effect of angled pressure side **1425** of annular ring **1130**, additional

acceleration is also provided by the converging nozzle configuration of inlet portion **1120** of outer housing **1105**. Accordingly, the fluid flow lines downstream of annular ring **1130** show further acceleration, which also diminishes with distance radially inward from outer housing **1105**. Accordingly, this combination of a converging nozzle of outer housing **1105** and a tilted airfoil shape of annular ring **1130** produces a significant amount of fluid acceleration, the greatest of which is in the radially outwardmost portion of the fluid flow cavity.

(88) The fluid acceleration provided by the converging nozzle of outer housing **1105** in combination with annular ring **1130** being the greatest in the radially outwardmost region is beneficial when this system is further combined with turbine wheel **1150**, since it is the turbine blades that are positioned radially outward and driven by the fluid flow that has been pre-accelerated by the converging nozzle of outer housing **1105** and annular ring **1130**.

(89) FIG. **18** is a schematic illustration of a cutaway cross-sectional view of turbine wheel **1150**. FIG. **18** illustrates exemplary orientations of turbine blades **1515** and compressor blades **1520**. In addition, FIG. **18** also illustrates the effect that turbine wheel **1150** has on fluid flowing through turbine blades **1515** and compressor blades **1520**.

(90) As shown in FIG. **18**, turbine wheel **1150** has an inlet side **1801** and an outlet side **1802**. Accordingly, FIG. **18** illustrates an upstream direction **1803** and a downstream direction **1804**.

(91) Turbine blades **1515** may have any suitable airfoil shape. In some embodiments, turbine blades **1515** may have an airfoil shape configured as described above with respect to FIGS. **1-5**. That is, turbine blades **1515** may have a hooked cross-sectional shape, as shown in FIG. **18**.

(92) As shown in FIG. **18**, turbine blades **1515** may be angled with a suction side **1870** of the airfoil cross-sectional shape being tilted downstream toward outlet side **1802** of turbine wheel **1150** and a pressure side **1875** of the airfoil cross-sectional shape being tilted upstream toward inlet side **1801** of turbine wheel **1150**. For example, as shown in FIG. **18**, turbine blades **1515** may each have a leading edge **1855** and a trailing edge **1860**. An axis **1865** through leading edge **1855** and trailing edge **1860** defines a line, which illustrates the tilt angle of turbine blades **1515**.

(93) Compressor blades **1520** may have any suitable airfoil shape. In some embodiments, compressor blades **1520** may have an airfoil shape configured as described above with respect to FIGS. **1-5**. That is, compressor blades **1520** may have a hooked cross-sectional shape, as shown in FIG. **18**.

(94) As shown in FIG. **18**, compressor blades **1520** may be angled with a pressure side **1892** of the airfoil cross-sectional shape being tilted downstream toward outlet side **1802** of turbine wheel **1150** and a suction side **1895** of the airfoil cross-sectional shape being tilted upstream toward inlet side **1801** of turbine wheel **1150**. For example, as shown in FIG. **18**, compressor blades **1520** may each have a leading edge **1880** and a trailing edge **1885**. An axis **1890** through leading edge **1880** and trailing edge **1885** defines a line, which illustrates the tilt angle of compressor blades **1520**.

(95) The turbine blades and compressor blades may be angled as described above at any suitable angles. In some embodiments, the turbine blades and the compressor blades may be angled such that a line extending through a leading edge and a trailing edge of a turbine blade is parallel to a line extending through a leading edge and a trailing edge of a compressor blade located radially inward from the turbine blade.

(96) FIG. **18** illustrates the effect of turbine wheel **1150** on the flow of fluid through different regions of turbine wheel **1150**. As shown in FIG. **18**, an annular outer region **1810** is defined between outer ring **1505** and intermediate ring **1510**. Fluid flow through outer region **1810** of turbine wheel **1150** passes through turbine blades **1515**. In addition, an annular inner region **1815** is defined between intermediate ring **1510** and inner hub **1500**. Fluid flow through inner region **1815** of turbine wheel **1150** passes through compressor blades **1520**.

(97) More particularly, fluid flow introduced through outer region **1810** is represented by an arrow **1820**. This fluid flowing through outer region **1810** around turbine blades **1515** drives turbine wheel **1150** in a counter-clockwise direction **1840**. By converting the energy from the fluid flow to

rotational movement of turbine wheel **1150**, the speed of the fluid flow coming out the outlet side **1802** of outer region **1810** is slower than at inlet side **1801** of outer region **1810**, as illustrated by a smaller arrow **1845**.

(98) Fluid flow introduced through inner region **1815** is represented by an arrow **1825**. Because of the orientation of compressor blades **1520**, this fluid flow is pulled through inner region **1815** by compressor blades **1520**, and thus, propelled in downstream direction **1804** from outlet side **1802** of inner region **1815**. A larger arrow **1850** illustrates the increased speed of fluid flow exiting inner region **1815** as compared to smaller arrow **1825** indicating the air entering inner region **1815**.

(99) Thus, fluid flowing through outer region **1810** and inner region **1815** at the same speed will result in the fluid coming out of outer region **1810** being slower and the fluid coming out of inner region **1815** being faster. As shown in FIG. **18**, arrow **1820**, representing fluid flow through outer region **1810**, is flowing at an outer radius **1830** from central axis **1430**, whereas arrow **1825**, representing fluid flow through inner region **1815**, is flowing at an inner radius **1835**, which is smaller than outer radius **1830**. Because turbine blades **1515** are located radially further from central axis **1430** of turbine wheel **1150** than compressor blades **1520**, the fluid flowing through outer region **1810** generates a larger amount of torque than required to propel the fluid through inner region **1815**. This results in an overall acceleration of the total flow of air through turbine wheel **1150**.

(100) Accordingly, turbine wheel **1150**, if placed in a fluid stream, will accelerate the fluid stream. When combined with the outer housing and airfoil-shaped annular ring of fluid accelerator **1100**, the fluid stream is greater. As discussed above with respect to FIG. **17**, the fluid flow in the radially outer areas is accelerated the most by the outer housing and annular ring. Accordingly, this fluid in the radially outer areas has even greater energy to drive turbine blades **1515** of turbine wheel **1150**, which results in greater propulsion of fluid through inner region **1815**.

(101) These three fluid accelerating components (i.e., 1. an outer housing with a converging nozzle, 2. an airfoil-shaped annular ring within the outer housing, and 3. a turbine wheel as described above), each provide fluid acceleration individually, and provide greater fluid acceleration when combined with one another. Further, these fluid accelerating components may be implemented to accelerate fluids in a variety of environments. Above, an exemplary use at an inlet to a ram air turbine is discussed. Another exemplary use for such a fluid accelerator is at the inlet of a jet engine.

(102) FIG. **19** is a schematic illustration of an airplane having fluid accelerators mounted at the inlets of its turbine engines. As shown in FIG. **19**, an airplane **1900** may include one or more jet engines **1905**. The inlet of one or more of such jet engines may include a fluid accelerator **1910**, which may have the same or similar configuration as fluid accelerator **1100** discussed above. Such an implementation of a fluid accelerator may pre-accelerate air entering the jet engines, enabling the engines to generate more thrust.

(103) While various embodiments of the invention have been described, the description is intended to be exemplary, rather than limiting, and it will be apparent to those of ordinary skill in the art that many more embodiments and implementations are possible that are within the scope of the invention. Any element of any embodiment may be substituted for another element of any other embodiment or added to another embodiment except where specifically excluded. Accordingly, the invention is not to be restricted except in light of the attached claims and their equivalents. Also, various modifications and changes may be made within the scope of the attached claims.

Claims

1. A turbine wheel, comprising: an inner hub having a central axis, an outer ring concentric with the inner hub, and an intermediate ring concentrically disposed between the inner hub and the outer ring, a plurality of turbine blades extending between the intermediate ring and the outer ring; the

turbine blades being oriented to be driven by a stream of fluid flowing through the turbine wheel to rotate the turbine wheel in a first direction about the central axis; and a plurality of compressor blades extending between the inner hub and the intermediate ring, each compressor blade including a leading edge and a trailing edge; the compressor blades being oriented to propel the stream of fluid in a direction downstream of the trailing edges of the compressor blades and axially with respect to the central axis when the turbine wheel is rotated in the first direction.

2. The turbine wheel of claim 1, wherein the turbine wheel has an inlet side and an outlet side; wherein the turbine blades have an airfoil cross-sectional shape including a suction side and a pressure side; and wherein the turbine blades are angled with the suction side of the airfoil cross-sectional shape being tilted downstream toward the outlet side of the turbine wheel and the pressure side of the airfoil cross-sectional shape being tilted upstream toward the inlet side of the turbine wheel.

3. The turbine wheel of claim 2, wherein the compressor blades have an airfoil cross-sectional shape including a suction side and a pressure side; and wherein the compressor blades are angled with the pressure side of the airfoil cross-sectional shape being tilted downstream toward the outlet side of the turbine wheel and the suction side of the airfoil cross-sectional shape being tilted upstream toward the inlet side of the turbine wheel.

4. The turbine wheel of claim 3, wherein the turbine blades and the compressor blades are angled such that a line extending through a leading edge and a trailing edge of a turbine blade is parallel to a line extending through a leading edge and a trailing edge of a compressor blade located radially inward from the turbine blade.

5. The turbine wheel of claim 1, wherein one or more of the turbine blades has a leading edge, a trailing edge, a suction side, a pressure side, and an airfoil cross-sectional shape including: a base portion including a first surface associated with the pressure side and a second surface associated with the suction side; an overhang portion that extends over at least a portion of the base portion; an elliptic portion connecting the base portion and the overhang portion adjacent the leading edge; and wherein the overhang portion is curved toward the second surface of the base portion.

6. The turbine wheel of claim 5, wherein the overhang portion comprises a first arc portion having a first radius of curvature on the suction side, the overhang portion further comprising a second arc portion having a second radius of curvature on the suction side that is different from the first radius of curvature.

7. The turbine wheel of claim 6, wherein the second radius of curvature is greater than the first radius of curvature.

8. The turbine wheel of claim 5, wherein a free end of the overhang portion is separated from the base portion by a gap, and wherein the gap is greater than a local thickness of the overhang portion.

9. A turbine wheel, comprising: a plurality of turbine blades oriented to be driven by a stream of fluid flowing through the turbine wheel to rotate the turbine wheel in a first direction about a central axis of the turbine wheel; and a plurality of compressor blades oriented to propel the fluid in a downstream direction when the turbine wheel is rotated in the first direction; wherein the turbine blades and/or the compressor blades have an airfoil cross-sectional shape, including: a base portion including a first surface associated with a pressure side and a second surface associated with a suction side; an overhang portion that extends over at least a portion of the base portion; and an elliptic portion connecting the base portion and the overhang portion adjacent a leading edge; wherein the overhang portion is curved toward the second surface of the base portion.

10. The turbine wheel of claim 9, wherein the overhang portion comprises a first arc portion having a first radius of curvature on the suction side, the overhang portion further comprising a second arc portion having a second radius of curvature on the suction side that is different from the first radius of curvature.

11. The turbine wheel of claim 10, wherein the second radius of curvature is greater than the first radius of curvature.

12. The turbine wheel of claim 9, wherein a free end of the overhang portion is separated from the base portion by a gap, and wherein the gap is greater than a local thickness of the overhang portion.
13. The turbine wheel of claim 12, wherein the gap is at least twice as large as the local thickness of the overhang portion.
14. A fluid accelerator, comprising: an outer housing having an inlet end and an outlet end, the outer housing defining a nozzle proximate the inlet end; and an annular ring disposed proximate the inlet end of the outer housing within the nozzle; wherein the annular ring has an airfoil cross-sectional shape; a turbine wheel disposed within the outer housing downstream from the annular ring, the turbine wheel including: an inner hub having a central axis, an outer ring concentric with the inner hub, and an intermediate ring concentrically disposed between the inner hub and the outer ring, a plurality of turbine blades extending between the intermediate ring and the outer ring; the turbine blades being oriented to be driven by a stream of fluid through the turbine wheel to rotate the turbine wheel in a first direction about the central axis; and a plurality of compressor blades extending between the inner hub and the intermediate ring; the compressor blades being oriented to drive the fluid downstream when the turbine wheel is rotated in the first direction.
15. The fluid accelerator of claim 14, wherein the nozzle is a converging nozzle.
16. The fluid accelerator of claim 14, wherein one or more of the turbine blades and/or one or more of the compressor blades each have a leading edge, a trailing edge, a suction side, a pressure side, and an airfoil cross-sectional shape including: a base portion including a first surface associated with the pressure side and a second surface associated with the suction side; an overhang portion that extends over at least a portion of the base portion; an elliptic portion connecting the base portion and the overhang portion adjacent the leading edge; and wherein the overhang portion is curved toward the second surface of the base portion.
17. The fluid accelerator of claim 16, wherein the overhang portion comprises a first arc portion having a first radius of curvature on the suction side, the overhang portion further comprising a second arc portion having a second radius of curvature on the suction side that is different from the first radius of curvature.
18. The fluid accelerator of claim 17, wherein the second radius of curvature is greater than the first radius of curvature.
19. The fluid accelerator of claim 16, wherein a free end of the overhang portion is separated from the base portion by a gap, and wherein the gap is greater than a local thickness of the overhang portion.
20. The fluid accelerator of claim 19, wherein the gap is at least twice as large as the local thickness of the overhang portion.
